

Three-Layer Evaluation Framework

No standard benchmark exists for DeFi risk monitoring agents -- we propose a layered design

Methods

m6_fm_llm	FM + LLM	Proposed
m5_fm_rules	FM + Rules	Ablation
m1_persistence_rules	Rules	Baseline
m2_snapshot_llm	LLM	Ablation
m4_fm_only	FM only	FM baseline
m3_evolvegcn	Temporal-GNN	Temporal-GNN

Layer 1: FM Prediction Quality

m5_fm_rules m1_persistence_rules m4_fm_only m3_evolvegcn

"Can the FM produce accurate graph forecasts?" [Runs on GPU]

b1_forecast: Risk Forecasting
MAE, Spearman rho, trend consistency

b2_warning: Early Warning
Lead time, precision, recall on 3 crises

b3_calibration: Uncertainty
ECE, PI coverage, CRPS

b6_robustness: Robustness
b1_forecast under 5 degradation regimes

Necessary condition

Layer 2: LLM Reasoning Quality

m6_fm_llm m2_snapshot_llm

"Given FM data, does LLM reason better than rules?" [Runs on CPU]

Groundedness
Fraction of cited values traceable to FM input

Faithfulness
LLM-as-Judge 1-5: reasoning follows evidence?

Consistency
Jaccard across 3 runs at temperature=0

Necessary condition

Layer 3: End-to-End Decision Quality

m6_fm_llm m5_fm_rules m1_persistence_rules m2_snapshot_llm

"Which method produces the best risk management decisions?" [Core contribution]

b5_decision: Decision Quality
Precision, recall, severity rho, stability, FIR, cost-adjusted

b4_stress: Stress Test
Contagion loss MAE, overlap@10

Explanation Quality
LLM-as-Judge 1-5 (rules = N/A by design)

Three claims to prove:

- (1) m6_fm_llm.Precision > m5_fm_rules.Precision (LLM reasoning > rule engine)
- (2) m6_fm_llm.Groundedness > m2_snapshot_llm.Groundedness (FM data reduces hallucination)
- (3) m6_fm_llm.Explanation > 0, Rules = N/A (LLM provides interpretability)